Practical No.8

MOHAMMAD QASIM

30-08-2025

Objective:

Write the program for finding the pseudo inverse of a matrix.

Python Code

In [2]:
```
import numpy as np

def PSI(M, tol=1e-10):
    detM = np.linalg.det(M)
    if abs(detM) > tol:   ## invertible if clearly nonzero ###
        AI = np.linalg.inv(M)
    else:
        SVD=np.linalg.svd(M)
        U=SVD[0]
        S=SVD[1]
        VT=SVD[2]
        sp=np.zeros(len(S))
        for i in range(len(S)):
            if S[i] < 1e-10:        ## very small singular values treated as zero ###
                sp[i] = 0
            else:
                sp[i] = 1 / S[i]
        S1 = np.diag(sp)
        AI=VT.T@(S1)@(U.T)
    return(AI)
```

In [3]:
```
M=np.matrix([[3,5,6],[6,10,12],[1,5,8]])
```

In [4]:
```
PSI(M)
```

Out[4]:
```
matrix([[ 0.0740458 ,  0.1480916 , -0.30152672],
        [ 0.02671756,  0.05343511, -0.05725191],
        [-0.0259542 , -0.0519084 ,  0.19847328]])
```

In [5]:
```
np.linalg.pinv(M)
```

Out[5]:
```
matrix([[ 0.0740458 ,  0.1480916 , -0.30152672],
        [ 0.02671756,  0.05343511, -0.05725191],
        [-0.0259542 , -0.0519084 ,  0.19847328]])
```

In [6]:
```
B = np.array([[1, 2],
              [2, 4]])
```

In [7]:
```
PSI(B)
```

Out[7]:
```
array([[0.04, 0.08],
       [0.08, 0.16]])
```

Interpretation:

As we can see the pseudo inverse of a matrix can be find by above function PSI().